6/8/26

EXPERIMENT 7

FLOW CONTROL AT DATA LINK LAYER USING SLIDING WINDOW PROTOCOL

AIM: To write a program to implement flow control at Data Link layer using the Sliding Window Protocol, and to simulate the flow of frames from one node to another.

ALGORITHM:

Step 1: Start the program.

Step 2: Read the total No. of frames to be sent (N), the sender's window size (W), (optionally) a frame and nextSeq = 0.

Step 3: Initialize base = 0. (seq. no. oldest unacknowledged frame)

Step 4: Repeat while base < N:

a. Determine the current sender window as the range.

b. Send all frames with sequence numbers from nextSeq. upto (base + W - 1), incrementing nextSeq after each frame, since the window restricts the sender to at most W outstanding.

c. If the lost frame lies within the frames just sent:

* The receiver discards the lost frame & any subsequent out-of-order frames.
* The sender times out and retransmits all frames starting from the lost frame (Go. Back. N).

Else (No loss in this batch):

* The receiver accepts each frame in order and sends a cummulative acknowledgement (ACK) for it.

OUTPUT:

```
=== Sliding Window Protocol - Flow Control Simulation (Go-Back-N) ===
Enter total no. of frames to send : 8
Enter sender's window size : 4
Enter frame no. to simulate as LOST : 5.
---------------------------------------
Sender Window: [Frame 0 - Frame 3]
Node A → Sending Frame 0
Node A → Sending Frame 1
Node A → Sending Frame 2
Node A → Sending Frame 3
Node B → Received Frame 0, sending ACK 0
Node B → Received Frame 1, sending ACK 1
Node B → Received Frame 2, sending ACK 2
Node B → Received Frame 3, sending ACK 3
Node A → ACK 3 received, sliding window forward.
---------------------------------------
Sender Window: [Frame 4 - Frame 7]
Node A → Sending Frame 4.
Node A → Sending Frame 5.
        (Frame 5 LOST in transit)
Node A → Sending Frame 6
Node A → Sending Frame 7
Node B → Frame 5 missing. Discarding out-of-order
         frames after it.
Node A → Timeout! Resending window starting from Frame 5
---------------------------------------
```

* On receiving the ACK for the last frame sent, the sender slides its window forward: base = next_seq.

5. Repeat ~~until~~ Step 4 until all N frames have been transmitted.

Step 6: Stop.

PROGRAM:

```
def simulate(total_frames, window_size, lost_frame):
    base = 0
    next_seq = 0
    while base < total_frames:
        window_end = min(base + window_size - 1,
                         total_frames - 1)
        print("\n - - - - - - - - - ")
        print(f" Sender Window: [Frame {base} - Frame {
                                window_end}]")

        while next_seq < base + window_size and next_seq <
        total_frames:
            print(f" Node A --> Sending Frame {next_seq}")
            if next_seq == lost_frame:
                print(f" (Frame {next_seq} LOST IN TRANSIT)")
            next_seq += 1

        if lost_frame is not None and base <=
        lost_frame < next_seq:
            print(f" Node B --> Frame {lost_frame}
                  missing. Discarding out-of-order
                  frames after it")
            print(f" Node A --> Timeout! Resending
                  window starting from Frame
                  {lost_frame}.")
```

Sender Window : [Frame 4 - Frame 7]

Node A → Sending Frame 5
Node A → Sending Frame 6
Node A → Sending Frame 7
Node B → Received Frame 4, sending ACK 4
Node B → Received Frame 5, sending ACK 5
Node B → Received Frame 6, sending ACK 6
Node B → Received Frame 7, sending ACK 7
Node A → Ack 7 received. Sliding Window forward

All 8 frames successfully transmitted from Node A to Node B.

```
        next_seq = lost_frame
        lost_frame = None

    for f in range(base, next_seq):
        print(f"Node B → Received Frame {f}, sending ACK {f}")
        print(f"Node A → ACK {next_seq-1} received. Sliding window forward.")
        base = next_seq

    print("\n----------")
    print(f"All {total_frames} frames successfully transmitted from Node A to Node B.")

def main():
    print("=== Sliding Window Protocol - Flow Control Simulation (Go-Back-N) ===\n")
    total_frames = int(input("Enter total No. of frames to send: "))
    window_size = int(input("Enter sender's window size: "))
    lost_frame = int(input(f"Enter frame to simulate as LOST\n(0 to {total_frames-1}, enter -1 for no loss)"))

    if lost_frame == -1:
        lost_frame = None

    simulate(total_frames, window_size, lost_frame)

if __name__ == "__main__":
    main()
```

RESULT:

The program to implement flow control at the Data Link Layer using the Sliding Window Protocol was successfully written.